

Market Signals: Evidence on the Determinants and Consequences of School Choice From a Citywide Lottery

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We estimate school-choice preferences revealed by the rank-ordered lists submitted by more than 22,000 applicants to a citywide lottery for more than 200 traditional and charter public schools in Washington, D.C. The results confirm previously reported findings that commuting distance, school demographics, and academic indicators play important roles in school choice and that there is considerable heterogeneity of preferences. Higher and lower income choosers respond to academic quality measures, but respond to different indicators of quality. Simulations suggest segregation by race and income would be reduced and enrollment in high-performing schools increased if policy-makers were to relax school capacity constraints in individual campuses. The simulations also suggest that removing the lowest performing schools as choice options could further reduce segregation and increase enrollment in high-performing schools.

Keywords: *school choice, school lottery, open enrollment, public school alternatives, school quality, school segregation, rank-ordered logit, exploded logit, policy simulation, demand forecasting*

UNDERSTANDING what parents value when they choose a school for their children is critical for learning how choice-based competition would influence school quality and the sorting of students into schools. Knowing the answer to this question would also help policymakers design the rules that govern public school choice and predict the behavioral implications of policies that could influence enrollment dynamics. Such policies can include opening, closing, or relocating schools. Knowing which parents tend to respond to which policy change, and in turn how that affects the composition of the school and subsequent choosers' willingness to attend, one can model the long-term effects of policy changes as well as the immediate first-year effects.

The collection of school options policymakers provide to parents and the rules that govern choice, sometimes referred to as “choice

architecture” (Thaler & Sunstein, 2008), is often viewed as a way policymakers promote individual freedom to choose while nudging choosers toward socially desirable outcomes. These outcomes for school choice are typically social integration (by race and income) and improved academic performance. If choosers are motivated by convenience and nonacademic amenities, then market-based competition will presumably lead to more of those amenities. Conversely, if choosers value academic performance, then choice should lead to competition along that dimension of quality. A third possibility is that when parents choose a school, they are really choosing a peer group. This would have its own implication for policy. Small policy nudges, such as changing the way schools are advertised to the public, can induce a few choosers to switch, which would change the makeup of the schools, and ultimately lead to very different sorting equilibria than the

status quo. Armed with information about how parents choose schools, policymakers could modify the choice architecture to increase the chances of achieving any goals over the status quo.

In this article, we empirically estimate these preference parameters, explore how they vary among types of consumers, and use them to simulate aggregate demand under alternative policies. To do this, we take advantage of a unified school lottery system in Washington, D.C., that began in 2014. This system centralized an existing regime of school choice that dated back to 1996, when legislation authorized charter schools and choice among traditional district schools. By 2014, the charter sector had enrolled more than 43% of public school students in D.C. The new common school-choice lottery gave parents the opportunity to apply to any of the more than 200 public schools in the city (including both charters and traditional schools) through a centralized assignment process, submitting rank-ordered lists of their most preferred schools. This process has not only decreased the cost of applying to multiple schools but also decreased search and information costs by providing an online school shopping and application portal that contains a searchable index of profiles for participating schools.

We compiled a rich data set that describes both the schools and the choosers, which we merged with the data on how each chooser ranked each school. With these data, we estimated a rank-ordered logit model to infer the preference weights that led parents to rank schools the way they did. Our additional goal in this article is to show how these taste parameters can be used to model the effects of changes to the choice sets or choice rules on sorting and academic performance. For example, if D.C. were to close the lowest performing schools, what effect would that have on average achievement and how would choosers sort themselves? Here, we make simplistic assumptions about the way in which choice improves academic outcomes. A more sophisticated analysis of supply-side response to competition is outside the scope of this article. See Cullen, Jacob, and Levitt (2006); Hastings, Kane, and Staiger (2006); and Deming, Hastings, Kane, and Staiger (2014) for work in this area.

Theory and Prior Research

Theorists have proposed several models for consumer choice. These include elimination by aspects, satisficing, and various forms of utility maximization (Ben-Akiva & Lerman, 1985; Bettman, 1971; Simon, 1947; Tversky, 1972; Wright, 1975). Each of these approaches implies a set of rules that describe how decision makers process information on alternatives. The theory we adopt here is utility maximization subject to a constrained discrete choice set (Ben-Akiva & Lerman, 1985). This theory, discussed more below, posits that consumers see schools as bundles of attributes, valued as the sum of the attribute values, weighted by the intensity of preference for each attribute. In the random utility model that we adopt, we distinguish between observed and unobserved attributes, treating the latter as stochastic. We assume that the ranking of any pair of alternatives is unaffected by the presence of the other alternatives (“independence of irrelevant alternatives” [IIAs]), and test robustness of the empirical results to that assumption.

By assuming a utility function that can be expressed as a linear combination of school attribute values, we hypothesize that choosers trade off school attributes against each other. In particular, we group those school attributes into three broad categories: convenience, peers, and academic performance. The empirical work is aimed at testing the weights among attributes that represent these broad categories.

A common empirical approach to studying how parents choose schools is to ask them directly through surveys (Armor & Peiser, 1998; Collins & Snell, 2000; Kleitz, Weiher, Tedin, & Matland, 2000; Schneider, Marschall, Teske, & Roch, 1998; Vanourek, Manno, Finn, & Bierlein, 1998). Despite differences in school-choice contexts and framing of school attributes, these studies share common findings that parents’ stated preferences suggest school academic quality is the most important factor in exercising school choice. Self-reported preferences, however, can be subject to socially desirable response bias (Stein, Goldring, & Cravens, 2010). Even carefully worded surveys might not elicit the desired information because choosers may not be fully conscious of all the preferences and biases that

guide their own decisions (Arkes & Tetlock, 2004; Nosek et al., 2007; Quillian, 2008).

The alternative to self-reports is revealed preference. Researchers have used Internet searches to gauge parental interest in topics (Schneider & Buckley, 2002) as well as enrollment patterns (Armor & Peiser, 1998; Stein et al., 2010), housing markets (Black, 1999; Bayer, Ferreira, & McMillan, 2007; Imberman & Lovenheim, 2016), interdistrict choice (Rothstein, 2006), and requests for teachers (Jacob & Lefgren, 2007). In contrast to findings based on stated preferences, studies using revealed preferences find that households with higher levels of income, parents' education, and student achievement are more likely to exercise school choice and place more importance on schools' test scores and student body composition. The evidence on teacher requests suggests that parents with higher income value teachers' principal-reported ability to improve student *satisfaction*, whereas parents with lower income value principal-reported ability to raise student *achievement*.

The method we used in this article, estimation of random utility models, is based on parent requests made within a centralized school-assignment process. In these situations, parents have a menu of alternatives, must submit their rank-ordered preferences, and are assigned using an algorithm that tries to satisfy those preferences. Random utility models, which assume that a continuous utility function is used by choosers to value and then rank-order items in a choice set, allow us to infer the preference weights that best predict the pattern of submitted rank-ordered choices. Studies in Minneapolis (Glazerman, 1998), Charlotte, North Carolina (Hastings, Kane, & Staiger, 2005), and New Orleans (Harris & Larsen, 2015) all found that nonacademic factors such as distance and racial composition were key determinants of the schools that families chose, but the studies highlighted different features of consumer demand. The New Orleans study highlighted parental preferences for extracurricular offerings. They also found that lower income families had a weaker preference for school academics. The Charlotte–Mecklenburg study explicitly modeled preference heterogeneity via random coefficients and noted that preferences for school academics were stronger for choosers with

higher baseline achievement and higher neighborhood income.

Three additional empirical papers from Belgium (Wouters, 2015), England (Burgess, Greaves, Vignoles, & Wilson, 2015), and the Netherlands (Ruijs & Oosterbeek, 2014) use school-choice rankings as well and reach similar conclusions as the U.S. literature. For example, choosers in Amsterdam were found to prefer schools in close proximity to their homes and schools sought after by same-gender classmates, whereas indicators of school quality were inconsistent predictors of choice.

Many scholars have speculated on the role that school choice plays in increasing (or reducing) social stratification, but there has been less empirical research in this area. The theoretical divide is well characterized by a debate in the *Sociology of Education* between Alexander Astin (1992), on one hand, arguing that “choice would magnify the existing social stratification of schools,” and James Coleman (1992), on the other hand, arguing that choice would “replace the current stratification by income and race by a stratification based on students' performance and behavior.” Hypothesized mechanisms that may affect the way stratification plays out include information networks (Schneider et al., 1998) and social capital and information available to choosers, particularly as they vary by race and class (Bell, 2009; Schneider, Teske, & Marschall, 2000; Weiher & Tedin, 2002).

To some extent, these predictions might depend on the rules governing the choice process (Henig, 1994). Fiske and Ladd (2000) hypothesized that choice would increase stratification and provided an example of this with a case study of New Zealand's system of public school choice. A key component of that system was that slots were controlled by schools and that oversubscribed schools did not have to use a lottery to allocate spaces the way they typically do in the United States. However, we would argue that regardless of the rules, the extent of stratification depends on the underlying preferences of those doing the choosing.

This Study's Research Questions and Contribution

This article addresses the following research questions:

Research Question 1: What school attributes do D.C. families value most when choosing schools?

Research Question 2: How do these preferences vary by type of chooser?

Research Question 3: What do these preferences imply for sorting over time under different policies?

We hypothesize that choosers value three types of attributes: (a) convenience, as proxied by commute distance from home to school; (b) peers, measured in terms of percentage of students in the prospective schools who are from low-income families and the percentage who are of the same race/ethnicity as the chooser; and (c) academics, measured in terms commonly used in the city, such as accountability ratings and scores that use state test results and other indicators.

We further hypothesize that choosers vary in predictable ways. Specifically, we seek to test whether choosers vary by race/ethnicity or family income. Such differences could be key drivers in choice-induced stratification and sorting. Finally, the article seeks to test whether the aggregate effects implied by the individual choice parameters lead to increased or decreased segregation by race and income, and whether the choice equilibrium leads to greater or less convenience for parents.

This article furthers our understanding of parent preferences by adding a new city and new ways to parameterize the preference function, differentiating by age groups and chooser characteristics. Even more important, the article explores the implications of choice behavior by simulating the results of choice policies with special attention paid to stratification by race and income.

We show in this article that, consistent with prior literature, parents tend to value convenience (schools close to home, shorter commutes, schools with access to public transit), demographics (schools where applicants are not in a small racial/ethnic minority), and academics (better ratings on accountability indicators or higher proficiency rates, but not higher growth scores). To our knowledge, this study is the first to use student-level application data from the common lottery in Washington, D.C., a city with extensive choice available to its more than

80,000 public school students. We examined how these preferences varied by chooser race/ethnicity, income, and grade to which the child was applying. For example, White and higher income applicants to elementary school had stronger preferences about distance and school demographics than African American applicants.

To better understand how demand for school attributes translates into sorting of students under various school-choice policies, we predicted rank-ordered choices and simulated student assignments under different sets of rules and conditions. After each prediction, we calculated the degree of sorting by race and income, summarized by the Index of Dissimilarity (D-Index; Taeuber & Taeuber, 1976), which varies between zero (groups are evenly distributed across schools) and 100 (groups are perfectly segregated). We note that it is difficult to quantify the effect of choice on academic quality because neither do we model the supply-side response, nor do we know much about sorting by a chooser's ability because such measures were not available in our data. However, our simulations suggest that if capacity constraints were lifted in individual campuses, there would be less racial segregation and possibly less segregation by income.

This study has some limitations to keep in mind when interpreting the results. First, the study excludes families who chose not to choose. Families with students entering Grades K through 12 had a right to attend neighborhood schools, and families with students returning to their current school or following a school feeder pattern did not have to participate in the lottery. Therefore, our results for middle and high school cannot be generalized to all families in the city. However, we estimated preferences separately for those entering pre-K, for which lottery participation was mandatory and universal, and those entering kindergarten, where lottery participation was optional, and the parameter estimates were nearly the same. Second, the study assumes that residential location is exogenous. That is, a family decides where to live and then ranks schools, rather than the other way around. Hastings et al. (2005) noted the possibility that endogenous residential location could lead to overestimating the importance of distance. If families move closer to schools they prefer, the observed relationship between distance and

school choice may in fact be masking a preference for specific school attributes. They addressed this by taking advantage of a policy that abruptly changed neighborhood school boundaries, likely producing exogenous variation in the default school. Harris and Larsen (2015) note that in a pure choice system such as that of New Orleans, where there are no by-right schools, the housing decision does not carry option value for a neighborhood school, so locating near a school is only necessary for reducing commute time. They also note other features of the choice process that likely mitigate concerns about endogeneity, such as the effect of parental preference for short commutes to work, complexity of the combinations of school characteristics, and the uncertainty, presumably of parents at the time of their residential location decisions, about where schools will locate. We explore the endogeneity of location by exploiting the depth of school rankings. Specifically, we test the sensitivity of our results by the rankings depth of choices. Under the assumption that choosers move to be near a top choice, the ordering of their lower ranked alternatives cannot be as easily gamed and, therefore, are more likely to represent true preferences over attributes.

Third, there are important omitted variables. Most notably, we did not have measures of the applicants' academic ability, so we could not model that aspect of heterogeneity. That would limit our ability to gauge whether school choice might lead to stratification by ability, other than through income or race. Fourth, we assume that choosers honestly rank their preferences. We discuss this assumption later in the article, but the basic justification is that the assignment algorithm is strategy proof and was conveyed as such through extensive outreach to families who were using the common school-choice application. In the remainder of the article, we describe the data and methods, empirical findings, simulation results, and conclusions.

Context, Data, and Method

Washington, D.C., Unified School-Choice Lottery

To understand the data for this study, it helps to know about school choice in Washington, D.C., and the process that generated the parent

rankings of schools. The public charter school sector in D.C. has grown rapidly since its beginning in 1996, with particularly rapid growth occurring in the early 2000s, both in number and popularity of schools. Historically, charter schools with demand that exceeds capacity have held their own lotteries for admitting students. In 2014, D.C. centralized school admissions by implementing a common application and lottery system for nearly all public schools (both traditional district schools and nearly all charter schools), known as My School DC. The traditional public school system, the DC Public Schools (DCPSs), sets aside seats for parents who wish to apply from outside their neighborhood boundaries. Under the new system, any family wishing to send their child to a school other than their by-right school (neighborhood school or school they currently attend) must submit a common application, ranking up to 12 of their most preferred schools from a list of all district and charter schools offering the relevant grade. Spaces available to lottery applicants in each participating school and grade level are then matched to students using a random assignment mechanism.

The student-school matching process uses a deferred-acceptance algorithm (Abdulkadiroğlu, Pathak, & Roth, 2009; Roth, 2008). Under this algorithm, each applicant is randomly assigned one unique number that determines the queue orderings among each group of students applying to the same school and grade level. When the number of applications to a school and grade exceeds capacity, the algorithm allocates available spaces according to this randomly generated queue ordering. Each student is then temporarily matched to their highest ranked school for which space was allocated to them, whereas any spaces allocated to the student in less preferred schools is then made available to other students. The algorithm repeats this process, attempting to match each student to his or her highest ranked school until no further improvements to match rankings are possible. For schools ranked as more preferred than the school the student is ultimately matched to, the student is assigned a waitlist position according to his or her queue position among all other applicants to that school who were not matched to a more preferred option. These waitlist positions are sequentially converted to matches

when other students decline enrollment in their matched school.

Some schools participating in the Washington D.C. unified school-choice lottery also add priority factors. These include whether the student is (a) currently attending another campus of the charter school for which the student is applying, (b) has a sibling currently enrolled at the selected school, or (c) has a sibling that was matched to the same school but is in a different grade. In addition, some D.C. public high schools are specialized high schools that assign students a queue position according to academic merit. Within schools that honor these additional priorities, the lottery assigns applicants within their relevant priority status group before using the lottery mechanism to determine the relative queue positions of students within each status group.

Importantly, the algorithm used by the common lottery is designed to be strategy proof, which means it disincentivizes any ranking of school choices other than by the applicant's true valuations for each ranked school. This is because (a) the matching method does not prioritize applicants for a given school based on the applicant's ranking of that school and (b) the lottery-assigned priority by which students are matched to available spaces does not depend on or vary across the set of schools chosen. Instead, the algorithm will always produce matches at the student's highest ranked school for which there is capacity, after accommodating students randomly assigned a higher match priority. Therefore, any rankings of school choices that deviate from an applicant's true preferences have the potential to result in a less preferred match and, in the most favored scenario, will result in a match of equal preference. Because our analyses rely on applicant rankings of schools reflecting true preferences, more important than the strategy-proof properties of the algorithm is that applicants perceive it to be strategy proof and behave accordingly. Indeed, My School DC used consistent messaging to parents, which not only specified that schools should be ranked according to their true preferences for enrollment, but also conveyed the strategy-proof design of the algorithm, and specifically explained that the likelihood of being matched to a preferred school did not depend on its ranking on the application.

Data Sources

We combine data from several sources to analyze parental preferences for schools in Washington, D.C. Applicants' rank-ordered school choices come from applications submitted for the 2014 D.C. common lottery. We match applicants to enrollment records from SY 2013–2014 (during which time applications were submitted) and SY 2014–2015. The combination of applications and enrollment data also allows us to identify the set of DCPS and charter school options available to students outside the lottery process. This set of additional school options includes “in-boundary” schools, which students are eligible to attend based on each applicant's address of residence; currently attended schools, which students may return to the following year; and “feeder” schools, to which students can matriculate from the school they currently attend. The matched enrollment data also provide characteristics of applicants, including students' gender, ethnicity, English learner status, special education status, and two proxies for income status; free or reduced-price lunch (FRL) eligibility, which is based on family income being below 185% of the poverty line; and direct certification for free meals, which indicates the student is homeless, served by the district's foster care program, or qualifies for the public assistance programs. Direct certification is preferred to FRL in D.C. because it was more carefully measured by schools in the period covered by our data.

To analyze preferences for schools as a function of school-level (and school grade-level) attributes, we bring in data from multiple sources, which were observable by parents at the time of application. These are (a) information from school profiles displayed on the My School DC website during the application period, including school size, availability of nearby public transportation lines, math and reading proficiency levels, racial composition of students, and before- and after-school child care services; (b) additional school-level information from the 2013 D.C. equity reports, which are school profiles released by the state education agency and include measures of gains in student proficiency levels and achievement-related accountability measures; and (c) annual data on reported crimes near school locations that were calculated by the

research team. Except for reported crimes, the information captured by these data were available to families as they shopped for schools. The My School DC Web portal linked to a school profile website that allowed parents to see snapshot information as well as a linked website called Learn DC that provided more detailed school profiles. The variables we used from these data sources are described in more detail in the “Findings on Parent Preferences” section.

Finally, some additional pieces of data are brought in to capture applicant-specific attributes of schools. Using school addresses of applicants and schools, we account for distance in four different ways: straight-line distance, right-angle distance, shortest driving distance using the fastest route according to a popular navigation service, and commuting time using the same route and navigation service. We also created the percentage of each school’s student body sharing certain characteristics of each applicant, including ethnicity, FRL eligibility, direct certification status, and English learner status.

We also use Census block group level data from the American Community Survey (ACS) to supplement information on both neighborhoods in which the schools are located and the neighborhoods in which the applicants are located. Census block groups contain between approximately 600 and 3,000 individuals, so these variables capture the income, education, race, and ethnicity of the immediate area around each school and each applicant.

Descriptive Statistics

We focus our analysis on the entry grade levels, specifically students applying for prekindergarten or kindergarten to enter elementary schools, Grade 5 or 6 to enter middle schools, or Grade 9 to enter high school, which are the most common grade levels for which families submitted applications. We obtained data for all choosers, including those applying to nonentry grades, but focus only on entry grades, as we have defined them, which accounted for 72% of all applications. Nonentry grade applicants are likely to have unusual motivations and circumstances than entry grade applicants. Preferences for school attributes and the ways that consumers behave have been shown to differ by grade

level in Washington, D.C., as well as other cities (Valant, 2014). There are many reasons: before-school care programs may be more desirable for young children than secondary students and children, who may value different things than their parents, can become more influential in school-choice decisions as they mature. Therefore, we conduct our analyses separately for elementary, middle, and high school entry grades. In Table 1, we describe the sample of schools and present the unique number of schools in the choice set for applicants in each of these grade ranges, as well as the unique number of school choices across all grade levels. Over all grade levels, charter schools make up roughly 45% of the school-choice options available through the lottery. This excludes 11 charter schools that opted out of the common lottery assignment system (of 102 charter schools in total) and held their own admissions processes in the first year.

The number of choices available at the high school level is smaller because high school students in D.C., as in most school districts around the country, are concentrated in a smaller number of larger buildings than are younger students. In our data, there were 32 high schools available to rank, nearly evenly split between the charter and traditional school sector. In addition, of the 15 participating DCPS high schools, 6 are selective programs that give preference in the selection process based on merit in the same way that an applicant with an enrolled sibling would have preference. Selective admission among these schools could have deterred some applicants with a low likelihood of admission from ranking such schools, but we suspect that the schools are omitted from the rank-ordered list, instead of being ranked lower than a less preferred school. Therefore, it is reasonable to interpret the ordering within the ranked alternatives, even for high school applicants, as a meaningful revelation of preferences.

We restrict our estimation sample to include all lottery applicants with a valid D.C. address—roughly 98% of all submitted applications—to ensure that we observe the distances between chosen schools and the applicant’s residence. As shown in Table 2, our sample includes 10,600, 2,389, and 3,036 unique applicants in elementary, middle, and high school entry grades,

TABLE 1

School Characteristics by Entry Grade (Percentages of Schools)

	Schools with elementary entry grades (pre-K, K, 1)	Schools with middle school entry grades (5 and 6)	Schools with high school entry grade (9)
District or charter			
DCPS	40	31	53
Public charter school	60	69	47
Percent African American			
0–20	13	12	0
21–40	7	9	6
41–60	12	10	11
61–80	15	14	14
81–100	53	55	69
Percent White			
0–20	81	84	91
21–40	9	7	9
41–60	4	4	0
61–80	6	5	0
81–100	0	0	0
Percent Hispanic			
0–20	74	74	83
21–40	10	9	9
41–60	9	9	3
61–80	4	6	6
81–100	2	2	0
Number of schools	135	136	35

Note. Schools may serve grade levels in more than one category. DCPS = D.C. Public Schools.

respectively. The dramatically larger number of applicants for elementary entry grades is explained by the fact that DCPSs do not have any right-to-attend options for prekindergarten grade levels. Therefore, all D.C. residents hoping to enroll their prekindergartners in a DCPS or charter school must apply for admission through the school-choice lottery.

The mean applicant characteristics presented in Table 2 show little variation across grade levels in the composition of applicants, save for slightly higher and lower percentages of White and African American applicants, respectively, at the elementary level.

Along observable characteristics, the pool of lottery applicants is fairly representative of the population of public school students in the city. In Table 3, we compare the composition of 2014 lottery applicants for all grades (pre-K through

Grade 12) with that of students concurrently enrolled in all DCPS or charter schools during the application and lottery process.

We observe some characteristics of applicants only within the enrollment data and, therefore, are unable to determine these characteristics for students who were both (a) not enrolled in a DCPS or charter school in the school year during the application process and (b) did not enroll in any DCPS or charter school in the subsequent year. For characteristics used in the analyses, we impute values for this group of students based on characteristics of the neighborhood (Census block group) in which they live using the ACS data. Although differences between these groups are statistically significant for several characteristics, most of these differences are not substantively meaningful. None is greater than 3 percentage points.

TABLE 2

Applicant Characteristics by Entry Grade

	Elementary: pre-K and kindergarten	Middle school: Grades 5 and 6	High school: Grade 9	All grade levels
Female	0.50 (0.50)	0.50 (0.50)	0.53 (0.50)	0.51 (0.50)
Race/ethnicity				
Asian	0.02 (0.11)	0.02 (0.13)	0.02 (0.12)	0.01 (0.10)
African American	0.65 (0.45)	0.72 (0.44)	0.75 (0.42)	0.71 (0.43)
Hispanic	0.14 (0.31)	0.14 (0.34)	0.16 (0.35)	0.14 (0.32)
White	0.16 (0.34)	0.10 (0.29)	0.06 (0.23)	0.11 (0.29)
Multiple/Other	0.03 (0.16)	0.02 (0.14)	0.02 (0.11)	0.03 (0.14)
Free or reduced-price lunch	0.49 (0.29)	0.53 (0.50)	0.54 (0.47)	0.53 (0.39)
Direct certified	0.36 (0.45)	0.42 (0.49)	0.39 (0.48)	0.39 (0.46)
Limited English proficient	0.11 (0.29)	0.05 (0.22)	0.07 (0.25)	0.09 (0.27)
Special education status	0.06 (0.20)	0.13 (0.33)	0.13 (0.33)	0.10 (0.28)
Distance to in-boundary school (miles)	0.8 (1.2)	1.3 (1.5)	2.1 (2.1)	1.3 (1.6)
Distance to any ranked school (miles)	2.9 (2.4)	4.0 (2.7)	4.5 (2.7)	3.5 (3.1)
Number of applicants	10,600	2,389	3,036	22,368

Note. Cell values are subsample means with standard deviations in parentheses. The fourth column includes all applicants applying for grade levels pre-K3 through Grade 12.

Model and Estimation Strategy

Our estimation strategy employs a discrete choice framework and investigates parental preferences for school characteristics using a rank-ordered logit model (Beggs, Cardell, & Hausman, 1981), sometimes referred to as an exploded-choice logit (Chapman & Staelin, 1982; Punj & Staelin, 1978). We begin by considering an applicant's choice of school as a utility-maximizing decision. In particular, we assume applicants' valuations of schools are a function of observable characteristics of the school, which include characteristics that do and do not differ between applicants of the same school, and other unobservable factors, which we treat as random. An applicant's utility derived from a particular school is defined as

$$U_{ij} = X_j' \beta + Z_{ij}' \gamma + \varepsilon_{ij}, \quad (1)$$

where i and j index individual applicants and schools, respectively. X_j is a vector of observable attributes of school j , including measures of schoolwide academic performance and school neighborhood demographics. Attributes of each school that vary by applicant, such as the proximity of the applicant household to the school, are captured by the vector Z_{ij} . ε_{ij} is an idiosyncratic term determined by unobserved factors of applicant i 's preferences for school j . The probability that applicant i values school j above all other $J - 1$ school choices then depends on the distribution of ε_{ij} . Assuming an independent type-I extreme value distribution leads to the typical logit model expression for this probability:

TABLE 3

Comparison of Lottery Participants to All Enrollees

	Lottery applicants	Enrolled students	Difference (Applicants – Enrollees)
Female	0.51 (0.50)	0.50 (0.50)	0.01
Race/Ethnicity			
Asian	0.01 (0.10)	0.02 (0.12)	–0.00*
African American	0.71 (0.43)	0.73 (0.44)	–0.02**
Hispanic	0.14 (0.32)	0.14 (0.35)	–0.01*
White	0.11 (0.29)	0.09 (0.29)	0.02**
Multiple/Other	0.03 (0.14)	0.02 (0.13)	0.01**
Free or reduced-price lunch	0.53 (0.39)	0.55 (0.50)	–0.02**
Direct certified	0.39 (0.46)	0.42 (0.49)	–0.03**
Limited English proficient	0.09 (0.27)	0.09 (0.29)	–0.00
Special education status	0.10 (0.28)	0.13 (0.34)	–0.03**
Number of students	22,368	76,296	

Note. Numbers in columns 1 and 2 are means for each student indicator with standard deviations in parentheses.

* $p < .05$. ** $p < .01$. *** $p < .001$.

$$\Pr[U_{ij} > U_{ik}, \forall k \neq j] = \frac{e^{X'_j \beta + Z'_j \gamma}}{\sum_{k=1}^J e^{X'_k \beta + Z'_k \gamma}}. \quad (2)$$

Because we observe the ordered rankings of multiple schools by applicants, we can use applications from the unified lottery to infer each applicant's ordinal preferences among all ranked schools. Using this larger set of information shifts our focus to the probability of applicants possessing a particular rank ordering of valuations among school choices, which is simply the joint probability of a series of sequential choices. For example, given a rank ordering where J school choices are indexed according to the strict preference ranking of the applicant, this probability is simply the joint probability of (a) School 1 being preferred to all schools, (b) School 2 being preferred to all schools that are not School 1, (c) School 3 being preferred among all schools that are not School 1 or 2, and so on. That is,

$$\begin{aligned} L_i &= \Pr[U_{i,j=1} > U_{i,j=2} > \dots > U_{i,j=J} | \beta, \gamma] \\ &= \prod_{j=1}^{J-1} \frac{e^{X'_j \beta + Z'_j \gamma}}{\sum_{k=1}^J \delta_{ijk} e^{X'_k \beta + Z'_k \gamma}}, \end{aligned} \quad (3)$$

where $\delta_{ijk} = 1$ if $U_{ij} > U_{ik}$ and 0 otherwise. This joint probability leads to the log likelihood function expressed in Equation 4, where N is the number of independent applicants and J_i denotes that the number of ranked school-choice alternatives can vary by applicant:

$$\begin{aligned} \log L &= \sum_{i=1}^N \sum_{j=1}^{J_i} X'_j \beta + Z'_j \gamma + \\ &\quad \sum_{i=1}^N \sum_{j=1}^{J_i} \log \left[\sum_{k=1}^{J_i} \delta_{ijk} e^{X'_k \beta + Z'_k \gamma} \right]. \end{aligned} \quad (4)$$

Under this setup, we obtain maximum likelihood estimates of the choice parameters β and γ .

Robust standard errors are calculated for coefficient estimates, which are equivalent to clustering standard errors at the applicant level under rank-ordered logit estimation. In section “Findings on Parent Preferences,” we give estimation results.

Explanatory Variables

We estimated a rank-ordered logit model that tested five domains of school-choice factors: convenience, school demographics, school neighborhood, school academic indicators, and other school offerings. There are many different ways to measure these factors and to specify their relationship with consumer rankings. We estimated a benchmark model that represents the most defensible functional form and choice of measures, but tested the robustness of this model by varying the measures, scaling, and specifications one at a time and reestimating the logit. Benchmark results are presented and discussed first, followed by a summary of the sensitivity analysis.

Convenience is measured by commute distance from home to each school alternative, where commute distance is computed as right-angle distance. Right-angle distance is the absolute value of the difference between school and home x coordinates plus the absolute value of the difference between school and home y coordinates. It assumes that travel occurs along a grid that only has east–west and north–south roads. We also tried a variety of alternative measures of commuting distance and time and found that the results were not sensitive to the choice of a particular distance metric. This domain also includes indicators for whether the school was near a subway stop or whether it was on two or more bus lines (to differentiate major from minor bus routes).

School demographics include the percentage of students in a school who are of the same race/ethnic group as the chooser. We estimate a quadratic relationship between the percentage of same-race students and school rankings. We later investigate heterogeneous preferences for the racial composition of a school’s student body and allow the own-race coefficients to vary by race/ethnicity of the chooser. The other school demographics variable is the percentage of students who are low income, as measured by direct certification for public assistance. All percentages are

divided by 10 so the coefficients can be more easily interpreted as the change in utility associated with an increase of 10 percentage points.

Several *academic indicators* were available as public information to parents choosing schools, some of which were more readily accessible than others. The most accessible one is the school’s proficiency rate in the prior year, which was prominently displayed on each school’s profile page on the My School DC website during the lottery application period. Other academic indicators were accessible through external sources, such as the annual D.C. equity reports, some of which were linked from the My School DC school profile pages. One such indicator is the school’s median growth percentile score from the prior year, which we refer to as the growth score. The median growth percentile is explained to parents as “a model that measures academic growth by comparing groups of students with similar test-score history.” It is calculated using the Colorado Growth Model (Betebenner, 2007; Walsh & Isenberg, 2013) and is meant to account for students’ prior achievement. We rescaled proficiency rates and growth scores, such as percentages, to range from 0 to 10, so that coefficients can be interpreted as the effects of a 10-point increase. In addition to the academic proficiency and growth indicators, we use categorical ratings from the city’s two accountability systems. One pertains just to charter schools and has three tiers: a top, middle, and bottom tier, labeled 1, 2, and 3, respectively. Charter schools are known to advertise their status if they are Tier 1 schools. The other accountability rating is for all schools in the city—district and charter—and has four levels, in order from lowest to highest score on a continuous accountability rating: priority, developing, rising, and reward. A fifth category, called focus, is assigned to schools that have large achievement gaps between subgroups of students and is typically considered a low rating similar to priority.

School neighborhood characteristics include the crime rates, measured as the average number of violent crimes per month reported within 500 feet of the school’s address over a 2-year period. The same measure is used for property crimes. As alternatives, we also calculated crime rates using larger radii of 1,000 and 1,500 feet from the school, as well as defining the area over which we

TABLE 4

Rank-Ordered Logit Results, Benchmark Model

School characteristics	Applicants entering . . .		
	Elementary school	Middle school	High school
Convenience			
Distance (miles)	-0.094***	-0.057***	-0.021**
On at least two bus lines	0.098***	0.225***	0.043
On subway	0.113***	0.063	0.136*
School demographics			
Own-race percentage/10	0.061***	0.163***	0.172***
Own-race percentage/10 squared	-0.002	-0.016***	-0.001**
Percentage low income (direct certified)/10	-0.161***	0.149***	0.038
Academics—proficiency and growth			
Proficiency rate/10	0.085***	0.070**	0.143***
Median growth percentile/10 (math)	-0.008	-0.111***	-0.196***
Academics—charter school accountability rating			
PMF Tier 1 (charters only)	0.101	0.228	-0.292*
PMF Tier 2 (charters only)	-0.003	0.028	-0.019
PMF Tier 3 (charter only)—omitted category	-0—	-0—	-0—
Academics—state accountability rating			
1. Reward	0.061	0.383**	1.204***
2. Rising	-0.028	0.121	0.541***
3. Developing	0.011	0.138	0.296*
4. Priority—omitted category	-0—	-0—	-0—
Focus school: Subgroup gaps	0.007	-0.262*	0.114
Sample size (student–school combinations)	51,572	9,045	11,543

Note. -0— parameter is constrained to zero. Variables with “/10” can be interpreted as one unit representing 10 percentage points. PMF is “Performance Management Framework,” which is used to rate charter schools. Sample for each column is applicants to entry grades for the respective school level: Pre-K and kindergarten for elementary, Grades 5 and 6 for middle school, and Grade 9 for high school.

* $p < .05$. ** $p < .01$. *** $p < .001$.

calculated per-capita crime rates for each school as its Census tract and block group. Other school neighborhood variables are drawn from the ACS and include several measures of socioeconomic status (SES) of residents of the Census block group in which the school is located. In the benchmark model, we included the log of the median household income in the school’s Census block group. Alternative specifications add more variables, such as the prevailing education levels in the Census block group, race/ethnicity, poverty, welfare receipt, and other demographics. Many of these variables were correlated with each other, so we relied on a more parsimonious specification for the benchmark model.

School neighborhood characteristics are included in the model primarily as control variables, but it is important to note that policymakers have considerable control over these factors through site approvals, disposal of “surplus” city buildings that can be used for charter schools, and selective school closures. The neighborhoods in which most charter schools and many DCPS schools are found result from decisions by the council and mayor of D.C. as well as the District’s Public Charter School Board.

Finally, the *other school offerings* include average class size (number of students per core class), an indicator for whether the school requires students to wear a uniform, availability

TABLE 5

Magnitude of the Logit Coefficient Estimates

Explanatory variable	Logit coefficient β	OR $\exp(\beta \times \Delta X)$	Change in probability of choosing over otherwise identical school $OR / (1 + OR) - 0.50$	Distance between two schools to make chooser indifferent $\beta/\beta \text{ (distance)}$
Distance: One-mile increase	-.057	0.94	-0.014	1.00 mile
Own race/ethnicity	.163			
Own race/ethnicity squared	-.016			
Increase from 10% to 20%		1.12	0.029	2.07 miles
Increase from 40% to 50%		1.02	0.005	0.36 miles
Proficiency rate: 10-point increase	.070	1.07	0.017	1.23 miles
Tier 1 versus Tier 3 on PMF	.228	1.26	0.057	3.86 miles

Note. Logit coefficients are from Table 4, middle school model. PMF is “Performance Management Framework,” which is used to rate charter schools. OR = odds ratio.

of before-school care, and availability of after-school care. All these variables are defined for DCPS schools only, not for charter schools. For charter schools, the values of each amenity variable are set to zero and a sector dummy is included. Sector is not the focus of this article, but other work has focused on sector in portfolio districts (Butler, Carr, Toma, & Zimmer, 2013; Lincove, Cowen, & Imbrogno, 2016). Other dummies are also included in the model, but not reported here because they do not have a useful policy interpretation. They include indicators for missing values that have been imputed and a dummy for whether the applicant has a sibling enrolled in the school.

Findings on Parent Preferences

Main Regression Results

The logit results are consistent with findings in the literature we summarized earlier. These results suggest that convenience, school academic performance, and student body composition are predictive of how parents rank school alternatives. The results in Table 4, showing separate models by entry grade, indicate that choosers were significantly more likely to rank schools higher if the schools were closer to home, on bus or subway lines; had a higher percentage of students from their own race/ethnic group; and had higher test score proficiency rates.

There is evidence at the middle and high school level that own-race preferences are nonlinear, on average. Specifically, choosers were more likely to rank a school higher if their own child’s race/ethnic group was better represented in the school, but that increase in likelihood of ranking the school higher was largest when the own-race/ethnicity percentage was relatively low. As the own-race/ethnicity percentage rises, the relationship is weaker and even negative, suggesting a taste for diversity.

The logit coefficients reported in the table are not literally the effect on school ranking probabilities. They represent the effect of each variable on utility in our model, but higher utility translates to a greater probability of being ranked higher, so the discussion here focuses primarily on sign and statistical significance of the coefficients. Standard errors for all logit coefficients are reported in the supplemental material, available in the online version of the journal.

Although we focus on the sign and significance of the parameter estimates, a helpful way to interpret the magnitude of the logit coefficients is to convert them to odds ratios, which express the relative odds of ranking the school higher than an alternative. In Table 5, we present some illustrative contrasts between pairs of middle schools that are identical in every way except for a change in one attribute. The distance coefficient of -0.057 implies an odds ratio of 0.94, which means that for every additional mile of commute distance to a

school, the relative odds of ranking it above the alternative fall by 6%. This can be translated further into a probability of choosing the school that was one mile farther over an otherwise identical school, which would fall from 50% (indifferent between the two) to 48.6%, a decline of about 1.4%. The own-race coefficients (on the linear and squared term) suggest that an increase in the percentage of students in the school whose race/ethnicity is the same as the applicant's from 10 to 20 percentage points is associated with an odds ratio of 1.12 for middle school choosers, whereas a contrast between 40 and 50 percentage points is associated with a smaller odds ratio of 1.02. For a 10-point increase in the percentage of students who are proficient on the state standardized test, the odds ratio is 1.07. For a charter school being in Tier 1 versus Tier 3, the odds ratio is 1.26, making the predicted probability of selecting it over an identical alternative 56%.

Yet another way to interpret the magnitudes of the coefficients is to express them in distance units. This is done by dividing any given coefficient by the coefficient on distance. For example, the model results suggest that parents would be willing to have their children commute an additional 1.23 miles to a middle school that has a 10-point higher proficiency rate on state standardized tests.

The benchmark model findings were almost all consistent across choosers applying to the entry grades for elementary school (prekindergarten or kindergarten), middle school (applying to Grade 5 or 6), and high school (applying to Grade 9). However, there were some differences in magnitude of effects, and within the domain of academic performance the specific indicator that was statistically significant varied by grade span. Although proficiency rate—the percentage of students scoring above a predetermined cut point on the state standardized test—was positively related to school preference for applicants of all grade spans, the categorical state accountability ratings were progressively more important for applicants of later grade levels. For applicants to elementary schools, the coefficients for these categories were not statistically significant. For applicants to middle school, the coefficient for the top accountability category was positive and significant, whereas each of the coefficients for the three highest categories was larger still among high school applicants.

The coefficient on the growth measure was negative and significant for both middle and high school applicants. One possible explanation for this is that with prior-year (2013) proficiency rate included in the model, growth is likely to capture lagged proficiency (2012) with the sign reversed. In other words, controlling for proficiency, schools with high growth are likely to be those that had lower proficiency in 2012. We did not include 2012 proficiency in the benchmark model because of collinearity with 2013 proficiency rates, but when we included it in an alternative model, the coefficient on school growth measures reversed sign for high school choosers. For middle school choosers, it did not change much at all, so this explanation did not consistently account for the observed findings.

We also examined preferences for school neighborhood characteristics, such as crime rates and median income, as well as amenities such as class size and school uniform policies. These are discussed in the Appendix, available in the online version of the journal.

Subgroup Analysis: Preferences by Income and Race

We model heterogeneity in consumer preferences in two ways. First, we estimate separate parameters for subgroups by interacting chooser characteristics with explanatory variables in the model. Second, we allow the coefficients to be random and estimate the moments of their distributions using a mixed logit model, discussed below in the sensitivity analysis.

Income. We first interacted all the model parameters with an indicator for low-income chooser. We found that the economic status of the choosers made a difference in preferences at the elementary and middle school levels, but not at the high school level. We summarize the results in Table 6. Among applicants to elementary schools, low-income choosers differed most from other choosers in terms of their preferences about school demographics. Notably, lower income choosers did not share the preference for schools with higher percentages of students of the same race/ethnicity and lower percentages of low-income students as higher income choosers.

TABLE 6

Interactions With Low-Income Status

School characteristics	Elementary school		Middle school		High school	
	Main effect	Interaction	Main effect	Interaction	Main effect	Interaction
Convenience						
Distance (miles)	−0.107***	0.037***	−0.042***	−0.032	−0.014	−0.017
On at least two bus lines	0.083***	0.038	0.298***	−0.179	0.073	−0.084
On subway	0.132***	−0.019	0.020	0.080	0.173*	−0.113
School demographics						
Own-race percentage/10	0.070***	−0.059*	0.189***	−0.064	0.142**	0.117
Own-race percentage/10 squared	−0.002	0.003	−0.019***	0.007	−0.007	−0.011
Percentage low income (direct certified)/10	−0.218***	0.190***	0.159***	0.001	0.033	0.045
Academics—proficiency and growth						
Proficiency rate/10	0.088***	−0.007	0.029	0.107*	0.130***	0.039
Median growth percentile/10 (math)	−0.007	−0.006	−0.091***	−0.026	−0.189***	−0.011
Academics—charter accountability rating						
PMF Tier 1 (charters only)	0.105	−0.015	0.378	−0.305	−0.168	−0.316
PMF Tier 2 (charters only)	−0.005	−0.031	0.131	−0.148	0.001	−0.011
PMF Tier 3 (charters only)—omitted category	−0—		−0—		−0—	
Academics—state accountability rating						
1. Reward	0.066	0.027	0.811***	−0.920***	1.405***	−0.431
2. Rising	−0.004	−0.017	0.378*	−0.511*	0.708***	−0.322
3. Developing	−0.003	0.038	0.235	−0.233	0.600***	−0.580*
4. Priority—omitted category	−0—		−0—		−0—	
Focus school: Subgroup gaps	0.064	−0.109	−0.130	−0.221	0.219	−0.152
Sample size (student–school combinations)	51,572		9,045		11,543	

Note. −0—no *p* value, parameter is constrained to zero. For each school level, main effect and interaction columns report coefficient estimates from the same regression. Coefficients reported in the interacted columns are those for the listed school characteristic interacted with the student’s direct certification status for free school meals. Because the coefficients reported in the main effects columns correspond to the overall association among all students, the sum of the main and interacted coefficients should be interpreted as the association between school characteristics and preferences for schools for direct-certified students. PMF is “Performance Management Framework,” which is used to rate charter schools. Sample for each column is applicants to entry grades for the respective school level: pre-K and kindergarten for elementary, Grades 5 and 6 for middle school, and Grade 9 for high school. Model includes school neighborhood and other school attributes, but coefficients not shown. Entry grades are pre-K and kindergarten for elementary, Grades 5 and 6 for middle school, and Grade 9 for high school.

p* < .05. *p* < .01. ****p* < .001.

Also, their distance preference was weaker by a statistically significant margin.

For applicants to middle school, the most interesting difference for low-income choosers is that, although their rankings were also influenced by school academic performance measures, the effect operated through the school’s proficiency rate and not the accountability categories. For

higher income choosers, however, it was the other way around.

These results have implications for how parents find and evaluate information on the academic quality of schools. Applicants for elementary school entry grades (pre-K and kindergarten) and lower income applicants to middle school more often ranked schools higher

based on proficiency rates—information that was directly observable on the My School DC website. Middle school applicants with higher income, and high school applicants in general, more often ranked schools higher based on accountability ratings—information that was not immediately available on the application portal and required more effort to find.

Race/Ethnicity. We also estimated interactions with each of the three largest race/ethnicity groups, as shown in Table 7. Those groups are White choosers, Hispanic choosers, and African American choosers. We grouped Asian choosers, which made up 1% of applicants, together with White choosers and we grouped those who identified as “multiple/Other” or missing, who together made up 16% of applicants, together with African American choosers. These groupings are arbitrary and did not change the conclusions when we put Asian, multiracial, and missing into a single omitted group of “Other.” We found differences among choosers’ race/ethnic groups in their own-group preferences and the nature of those differences varied, depending on the grade span to which the choosers were applying (elementary, middle, or high school). For each entry grade, the joint hypothesis test of whether the coefficients for African American choosers differed from those of each of the other two groups was easily rejected, meaning that differences were significant overall. Many of these preference parameters by race/ethnicity subgroup are no longer statistically significant once we disaggregate. Nevertheless, we present the results from Table 7 in graphical form in Figure 1, as exploratory findings, which may provide clues about race preferences that should be explored in greater depth in contexts where there is more overlap in racial composition of the school alternatives. The confidence bands around each curve indicate the degree of uncertainty in estimating these relationships. Most notably, they imply that we should be most cautious about drawing conclusions on the own-race effects for White choosers.

In D.C., there were large differences in the range of own-race variation each group faced. African American choosers were typically deciding between schools where they might be in a majority versus a super majority, whereas White and Hispanic choosers were typically deciding

between schools where they would be in a small minority versus a larger minority.

For those applying to enter elementary school, White choosers tended to prefer schools with greater percentages of students who were from their same race/ethnic group. Also, for White choosers, the preference was strongest for schools where their children would be in a small majority (up to 60%), with a slight decline in this preference among the few schools with a higher proportion of White students, as evidenced by the curvature of the relationship between own-group percentage and utility shown in the top panel of Figure 1. For Hispanic choosers, the relationship was more nearly linear, although these coefficients were not statistically significant (individually as coefficients on linear and quadratic terms or jointly). Meanwhile, African American choosers essentially showed indifference for own-group racial composition of schools.

The picture looked different for those applying to middle school entry grades (see Figure 1, middle panel), where all but the Hispanic group of applicants had a pronounced own-group preference and a slight preference for diversity, evidenced by curvature and a peak within the data at 50%. The bliss point, or optimal own-group percentage, was 26% for White applicants to middle school and 47% for African American applicants. In each figure, the curves are intentionally truncated, to avoid extrapolating beyond the range of observed own-group percentage for each group. The results for applicants to high school are shown in the bottom panel of Figure 1, which makes it clear that there was even less diversity of schools in terms of racial composition.

Although there were some statistically significant interactions with other variables (significance results not shown, but available from the authors), there were very few consistent patterns, and most of the preference parameters were of similar size and direction for all the race/ethnic groups. Exceptions at the elementary level were the demographic preferences, where coefficients for White choosers on own race/ethnicity and percentage of low income were significantly different from those of African American choosers. Coefficients for Hispanic choosers on percentage low income and the accountability ratings were also significantly different from the correspondent coefficients for African American choosers.

TABLE 7

Separate Model Parameters by Race/Ethnicity

School characteristics	Elementary			Middle school			High school		
	African American	White	Hispanic	African American	White	Hispanic	African American	White	Hispanic
Convenience									
Distance (miles)	-0.076***	-0.209***	-0.079***	-0.064***	-0.017	-0.047	-0.022**	-0.004	-0.031
On at least two bus lines	0.094***	0.031	-0.033	0.165***	1.610***	0.006	0.005	-0.931	0.345
On subway	0.130***	0.143***	0.215***	0.016	0.310	0.235	0.136	1.834*	0.233
School demographics									
Own-race percentage/10	-0.030	0.084**	0.042	0.188***	0.178	0.040	0.294	-0.169	0.485*
Own-race percentage/10 squared	0.002	-0.006	0.003	-0.020***	-0.034	-0.004	-0.019	0.003	-0.064*
Percentage low income (direct certified)/10	-0.069***	-0.456***	-0.165***	0.214***	-0.110	-0.037	0.072	-0.033	0.182
Academics—proficiency and growth									
Proficiency rate/10	0.090***	0.106***	0.056*	0.089**	0.144	-0.053	0.150***	-0.215	0.192*
Median growth percentile/10 (math)	-0.021**	0.014	-0.021	-0.093***	-0.251***	-0.125*	-0.222***	-0.248	-0.203**
Academics—charter accountability rating									
PMF Tier 1 (charters only)	-0.008	0.204	0.334*	0.184	2.122***	0.598	-0.333*	-0.438	-0.044
PMF Tier 2 (charters only)	-0.036	-0.045	0.045	-0.053	0.538	0.644	-0.125	-1.669	0.222
PMF Tier 3 (charters only)—omitted category	-0-	-0-	-0-	-0-	-0-	-0-	-0-	-0-	-0-
Academics—state accountability rating									
1. Reward	0.023	0.004	0.292	0.340*	-0.066	0.627	1.263***	5.257*	1.416*
2. Rising	-0.066	0.026	0.173	0.128	-0.213	0.385	0.589***	3.338*	0.882
3. Developing	-0.008	-0.288	0.416	0.134	0.908	0.001	0.270	1.611	1.198*
4. Priority—omitted category	-0-	-0-	-0-	-0-	-0-	-0-	-0-	-0-	-0-
Focus school: Subgroup gaps	-0.036	0.136	0.317*	-0.211	0.258	-0.104	0.230	1.316	0.331
Sample size (student–school combinations)		51,572			9,045			11,543	

Note. -0– no p value, parameter is constrained to zero. PMF is “Performance Management Framework,” which is used to rate charter schools. Sample for each column is applicants to entry grades for the respective school level who identify as the corresponding race category. Model includes school neighborhood and other school attributes, dummies for missing values, but coefficients not shown. Entry grades are pre-K and kindergarten for elementary, Grades 5 and 6 for middle school, and Grade 9 for high school.

* $p < .05$. ** $p < .01$. *** $p < .001$.

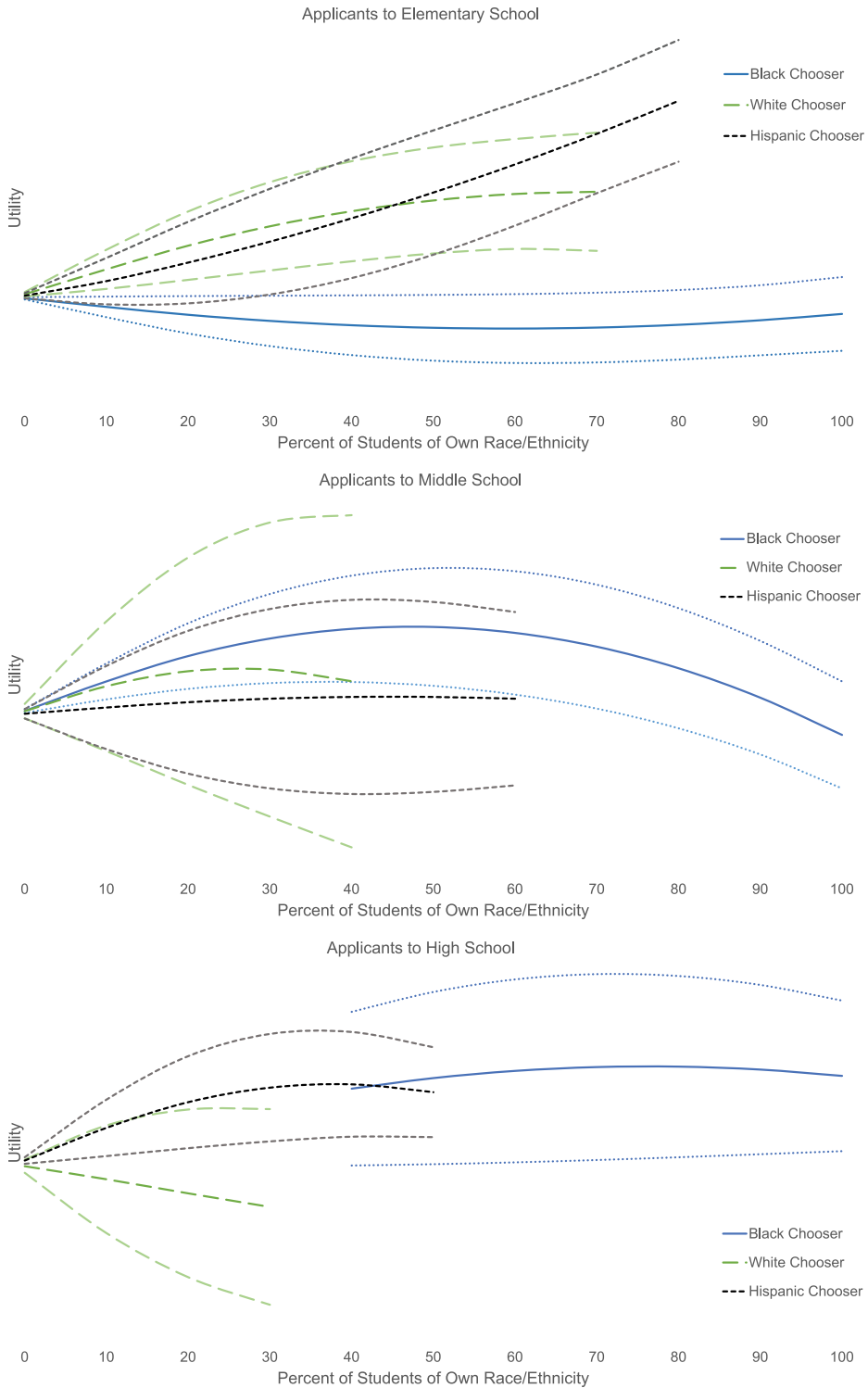


FIGURE 1. *Own-race/ethnicity preferences, by race/ethnicity of chooser.*

Note. For each chooser-type, three lines from top to bottom represent the 95 percent confidence interval (CI) upper bound, mean utility level, and 95 percent CI lower bound, respectively, for the corresponding percentage of students in the school sharing the chooser's own race/ethnicity category.

Robustness

We examined several alternative functional forms to test the robustness of the main findings. In most cases, the general conclusions did not change a great deal. However, allowing for more interactions and more flexible nonlinearities in the relationships between attributes and utility led to some more nuanced findings.

Distance. It is plausible that preferences for a closer school are not linear. For example, there may be a mode-switching argument that a chooser first decides whether the prospective school is within walking range and then if not, considers sending the child to school in a car, bus, or train. The time cost of an extra half mile walk may be considerably more than the time cost of riding in a car, bus, or train for an additional half mile. Therefore, we estimated a piecewise linear model, with switching over at one mile. The estimates, shown in the online appendix, suggest there is indeed a discrete penalty associated with switching from walking to riding, but that the per-mile penalty (slope of the relationship between distance and utility) decreases substantially beyond one mile. We also tested a quadratic model and models that used alternative measures of distance, including a straight-line measure and another that used mapping software to estimate the actual shortest route travel distance. The alternative distance measures were highly correlated with each other, so, not surprisingly, the estimated preference weights were similar as well. In the quadratic model, the distance-squared term was statistically significant, although the degree of curvature was modest over the typical range of zero to six miles.

Academics. The data suggest that a school's academic performance is a statistically significant determinant of school preference, but there are several ways to measure academic performance, none of which consistently explains this relationship. The results presented in the benchmark model used four kinds of measures: proficiency rates varying from 0% to 100%, median growth percentile scores ranging from 0 to 100 as well, and two types of categorical accountability ratings—charter and state ratings, as discussed above.

To test the robustness of the benchmark model, we estimated several alternative specifications that each included one measure of school academic performance at a time: proficiency rate, growth score, indicators for each charter accountability rating, and indicators for each state accountability rating. These are shown in Table 8 for applicants to middle schools, the grade span where standardized tests are given in every grade. Similar results for applicants to elementary and high school grades are shown in the online appendix.

The findings presented in Table 8 as well as the findings in the online appendix for other grade spans reinforce the conclusion that schools with better academic indicators are more likely to be ranked higher, although there were some exceptions. The statistically significant coefficient of .11 for proficiency rate was in line with the coefficients for elementary and high school (.09 and .18, respectively). For growth percentile score, however, the coefficient was negative for middle school applicants (−0.03, significant at the 5% level), but positive for elementary school applicants (0.03) and high school applicants (0.14). The accountability categories all had coefficients that followed the expected pattern of higher categories being preferred. The size of the premium placed on higher ratings was greater for upper grades than lower grades, suggesting that nonacademic factors weigh more heavily for parents of younger children. We also estimated models with more than one indicator together (as in the benchmark model) and separate models using continuous measures only or categorical measures (as dummy indicators). Those results are shown in the online appendix.

Endogeneity and IIAs. Parameter estimates from our approach provide suggestive information about the relative valuations families place on school attributes. However, there are some threats to validity to consider. To address endogeneity of distance, we re-estimated the benchmark model omitting each applicant's top two school choices, relying on the fact that if applicants do choose a residential location on the basis of schools, they would only do so on the basis of a top choice or second choice. The resulting estimates did not differ substantially from the main results, suggesting the potentially endogenous distance to schools is not a source of significant bias.

TABLE 8
Alternative Measures of Academic Quality: Middle School Entry Grades (5 and 6)

School characteristics	Proficiency rate	Median growth percentile	Charter tier	ESEA classification	Proficiency and ESEA classification	Charter tier and ESEA classification
Proficiency and growth						
Proficiency rate/10	0.11***					
Median growth percentile/10 (math)		−0.03*			0.06**	
Charter accountability						
PMF Tier 1 (charters only)			0.36***			0.23**
PMF Tier 2 (charters only)			0.21*			0.12
PMF Tier 3 (charter only)—omitted category			−0—			−0—
State accountability						
Reward				0.52***	0.32*	0.49***
Rising				0.18	0.05	0.18
Developing				0.29*	0.15	0.30*
Priority—omitted category				−0—	−0—	−0—
Focus				−0.25*	−0.32*	−0.24
Sample size (student–school combinations)	9,045		9,045		9,045	

Note. −0— no *p* value, parameter is constrained to zero. PMF is “Performance Management Framework,” which is used to rate charter schools. ESEA is the Elementary and Secondary Education Act, for which schools receive an accountability index score. These scores are used to classify schools in the discrete categories used in the relevant regression specifications. Model includes convenience, school demographics, school neighborhood, and other school attributes, but coefficients are suppressed here. Corresponding results for elementary and high school entry grades are shown in the online appendix.
 p* < .05. *p* < .01. ****p* < .001.

We also investigated the plausibility of the rank-ordered logit model’s implicit assumption of IIAs in the context of school-choice sets. IIA violations could lead to biased estimates. First, we randomly excluded ranked school choices and jointly tested differences between the resulting parameter estimates and the estimates of our benchmark model, generating a distribution of test statistics. We also show in the online appendix that the distribution of *p* values from these iterative tests suggest our likelihood of rejecting the null that the two set of instruments are jointly different is only slightly higher than random chance.

Second, we compare our benchmark model estimates using the conditional logit with those from a mixed logit model, which does not inherently assume IIAs. We find that the sign and

significance of coefficient estimates generally do not change when relaxing IIAs, with the exception of some statistically insignificant estimates that are close to zero. Together, these results suggest that if IIA violations do occur in the data, they do so relatively infrequently and are unlikely to introduce large errors in predicted choice probabilities.

Simulation Results: How Preferences Lead to Sorting
Measuring Sorting Outcomes Such as Stratification

We showed above that many attributes of schools influence how choosers rank their options. To better understand the implications of this pattern of preferences, we used the estimated

parameters to predict how parents would rank schools under alternative scenarios. By using our model's predicted probabilities of selecting schools, we calculated the resulting distribution of students across schools if we were to relax all capacity constraints and allow everyone to attend his or her most preferred school. The resulting allocations of students to schools were then compared with the allocation that resulted under the status quo, which assigned students to their top choices, subject to a deferred-acceptance lottery to handle oversubscription. The predicted allocations under unrestricted choice (no capacity constraints) and status quo were also compared with that of a policy of neighborhood schools and to alternative policies in which we place restrictions on the choice sets. These restrictions simulate the effects of closing low-performing schools, for example, by simply removing those schools from the choice sets and predicting choices from among the remaining alternatives. Our model allows us to answer the following question: If we close a low-performing school, where would the parents who would have selected that school choose to attend instead?

For each scenario, we defined outcome measures that summarize some aspect of the allocation of students to schools. One such outcome is the D-Index, which has been used to measure social stratification in a variety of contexts (Taeuber & Taeuber, 1976). The D-Index score for any binary measure, such as race, is defined as

$$D = \frac{1}{2} \sum_{j=1}^J \left| \frac{w_j}{W} - \frac{n_j}{N} \right| \times 100, \quad (5)$$

where w_j is the number of White students in school j , W is the number of White students in all schools, n_j is the number of non-White students in school j , and N is the number of non-White students in all schools. D ranges from a score of 0 for perfect integration to 100 for perfect segregation and can be interpreted as the percentage of students of a given group that would have to move to reproduce the overall fraction of each group in every individual school. We also examined whether students sorted themselves into higher performing schools by calculating the average enrollment (when capacity constraints are lifted) in the schools with the highest academic ratings. And finally, we examined how the

school-choice policies affected average commute times.

Simulation Results

We calculated D-Index scores for both race (White non-Hispanic and all Others) and for income (direct certified as eligible for FRL meals or not direct certified). Results are shown in Figure 2, with elementary, middle, and high school results in the top, middle, and bottom panel, respectively. The figures show that under the status quo, the assignment process populates schools with a D-Index of 65 for race and 36 for income at the elementary level. In other words, 65% of White students would have to be moved to achieve perfect balance by race, and 36% of direct-certified students would have to move to achieve perfect balance by income.

Going to neighborhood schools, which would assign all applicants to their nearest by-right DCPS school (which excludes charter schools) would reduce the D-Index for race from 65 to 60, but increase it for income from 36 to 40. If all applicants were offered their most preferred school listed in their applications without regard to capacity constraints (unrestricted choice), then the D-Index scores would be lower still, 57 for race and 36 for income. However, our model predicts that if we constrain choices by removing the lowest performing schools from the choice set, both D-Index scores decline to 48 and 34 for race and income, respectively. A similar pattern holds for middle and high school, where constrained choice (eliminating the lowest performing schools) produces the lowest levels of stratification of all the policy options considered, although not always very different from unrestricted choice.

We did not have information on choosers' ability levels and were, therefore, unable to examine the effects of different policies on sorting by ability. However, we did examine whether different policies would increase enrollment in schools that were higher versus lower performing in the year prior to the lottery. The analysis, summarized in Figure 3, shows that compared with assignments based on neighborhood catchment boundaries, school enrollments under the status quo lottery process result in many more middle and high school students attending schools with standardized testing proficiency

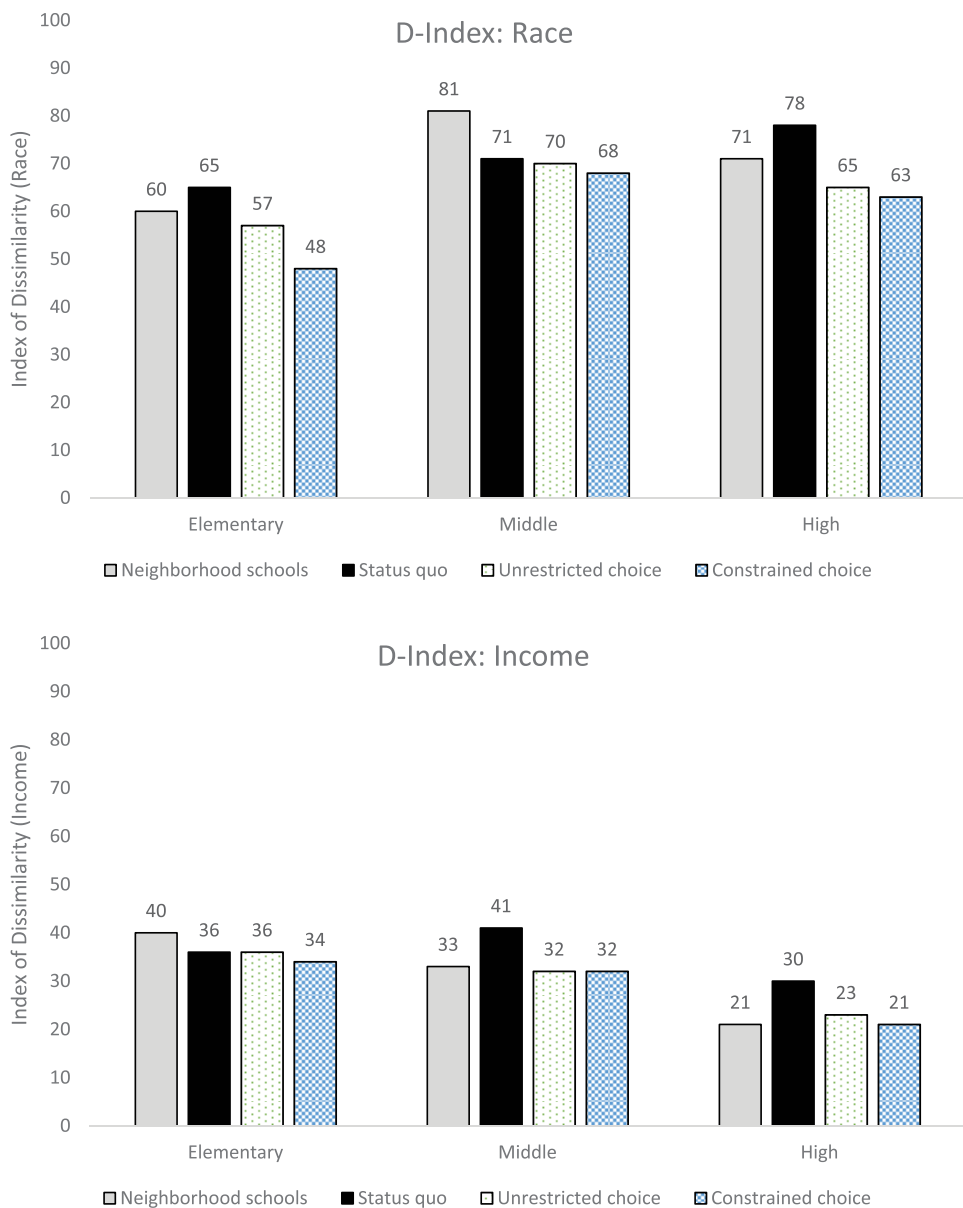


FIGURE 2. *Race and income stratification under alternative policies.*

rates in the top quartiles. For all grade levels, both unconstrained (predicted choices) and constrained choice (predicted choices in the absence of the lowest performing schools) result in even greater shares of students attending schools with higher proficiency rates. Interestingly, with the exception of elementary, the constrained choice predictions produce results very similar to unconstrained choice. Taken together, these simulation results suggest that for students in tested grades, expanding school-choice capacity by

making more spaces available at each campus would lead to additional movement of students toward schools with reputations for higher levels of academic achievement beyond that induced by the status quo school-choice process (although not necessarily higher levels of academic growth, as discussed in section “Findings on Parent Preferences”). The smaller differences between status quo, neighborhood schools, and unrestricted choice for elementary entry grades can be explained by the fact that some applicants

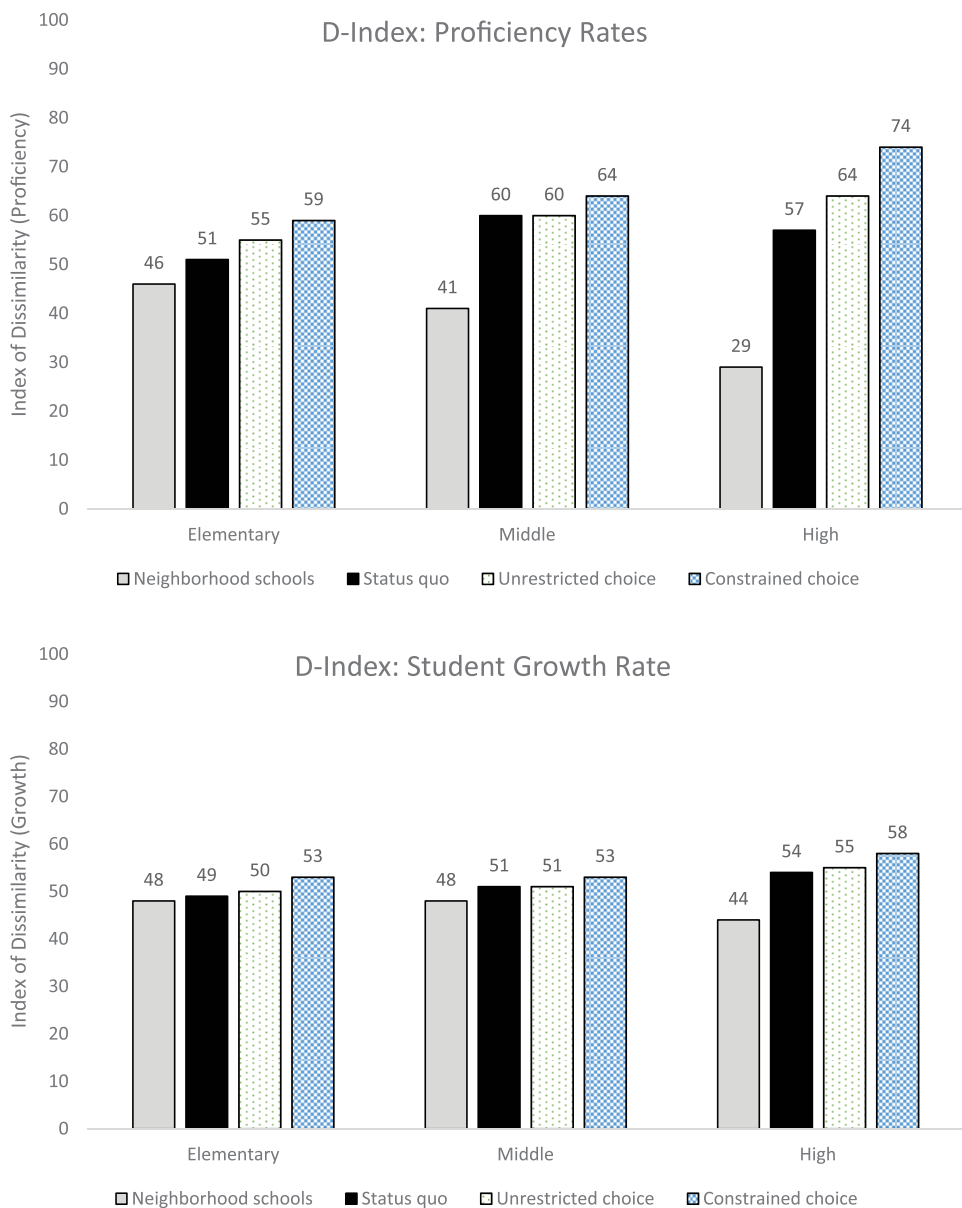


FIGURE 3. Enrollment-weighted school proficiency rate and growth score under alternative school-choice policies.

chose schools that serve only early grade levels and, therefore, do not receive standardized testing proficiency rates. In addition, parents may place a lower value on schools' academic achievement rates than on other attributes until their children are closer to entering a tested grade level.

Looking at commute times, we found that school choice resulted in longer commutes to school. Figure 4 shows the average commute

distance would be lowest if everyone attended an in-boundary school—less than a mile for elementary students, just more than a mile for middle schoolers, and less than two miles to high school. Under the status quo, participants in the choice process enrolled in schools that were an average of 2.5, 2.9, and 3.6 miles away. Our model predicts that consumers would prefer even longer commutes to get their most preferred school—3.8, 5.5, and 5.0 miles to elementary,

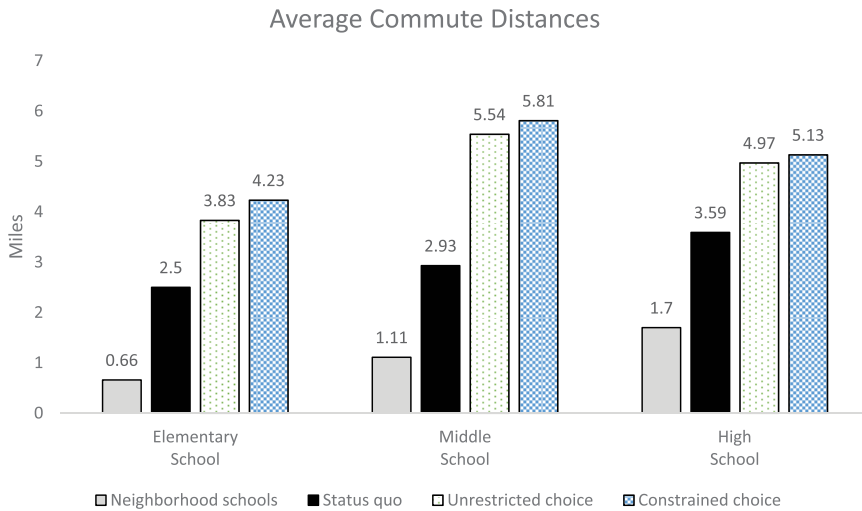


FIGURE 4. *Average commute distances to school under alternative school-choice policies.*

middle, and high school, respectively. Removing the lowest performing schools would result in even longer commutes, but not by much, less than three tenths of a mile farther on average.

The simulation methods used here can be extended in a number of other ways. One can place other restrictions on choice sets and examine resulting enrollment patterns from any number of perspectives, such as an individual school or school type, including enrollment by different subgroups of students. One can also simulate the effects over time. We examined the effect of updating the student body composition after predicting enrollment outcomes for unrestricted choice and repeating the prediction exercise until a steady-state equilibrium was reached. This steady-state analysis converged after approximately five iterations, with further iterations not generating substantially different predictions, so we did not report findings after five iterations here. In theory, however, long-run resorting can take place over time as choosers adjust to the changing demographic composition of the school (see Glazerman, 1998).

Contributions, Limitations, and Future Research

The current study shows how school application data can be used to better understand the preferences of parents when there is a marketplace of schools, and to predict the consequences

of these preferences for sorting and stratification. This is important because the number of U.S. cities that have adopted, are soon to adopt, or are considering adopting centralized application and assignment systems is growing rapidly (Rich, 2016). Part of this growth is stimulated by the expansion of charter schools and the need to rationalize the large numbers of parallel school admissions processes for parents and school administrators. With this growth, there will come more and better applicant data to understand consumer preferences.

A major contribution is the policy simulation as a way to reconcile the competing influences of different types of preferences. We hypothesized that parents trade off convenience with peers' characteristics and school academic performance. Although convenience and composition of the student body were important, the simulations showed that enrollment in high-performing schools went up under policies that expand choice, whereas average commute distance went up (i.e., convenience went down) and segregation, for the most part, was similar or lower than under neighborhood schools in most cases. This suggests that measured performance was a stronger influence, ultimately, than the other factors.

We also tested hypotheses about consumer heterogeneity, finding several differences among chooser types as well as age of the student (entering elementary, middle, or high school). Many of

these differences interact, for example, the heavier influence of qualitative school accountability ratings over test score proficiency rates, which was reflected at middle school entry grades for higher income choosers, but not at elementary or high school. Others may be more reflective of the available choice sets. African American choosers, for example, demonstrated an own-race preference that peaked at less than 100%, suggesting some taste for diversity, but this finding was only detectable at the middle school level, where there was a wide range of both segregated and integrated schools.

Some important qualifications and limitations apply to the findings from this study and others such as it. First, we necessarily focus on those who chose to participate in the lottery. In D.C., as in most settings, there is a default school or a secondary assignment process that takes place after the lottery deadlines have passed, in which large numbers of students would enroll in schools that have available space. Sorting outcomes may depend critically on who declines or fails to participate in the choice system, so a worthwhile focus of future research would be the factors that affect choice along this margin.

Another limitation is that we treat residential location as exogenous to the school-choice decision. A third limitation has to do with important unobserved variables. Neither did we have data on students' own academic ability, nor did we have information on choosers' attributes that might predict preference for special programs such as music, sports, or special services. Access to these variables, which were shown by Harris and Larsen (2015) to be relevant in New Orleans, could improve the fit and predictive power of the model and reduce the amount of unobserved idiosyncratic heterogeneity.

This article's finding that rankings of lower income choosers were explained by different indicators than higher income choosers is consistent with the theory of search costs, suggesting that there may be barriers to obtaining information about certain school attributes. However, we note that all the inferences made in this article are conditional on a particular static information regime. That is, we can describe what information was readily available to choosers and what information would have been more difficult to obtain, but cannot easily model how outcomes

would change under alternative information regimes. If policymakers were to change that balance, for example, by lowering the cost of learning about idiosyncratic school characteristics or academic quality, then those attributes might be more influential. A fruitful area for future research would be conducting information experiments, wherein policies are tested that might reduce search costs or the costs of learning about schools or particular school attributes.

In conclusion, the basis of competition in a school-choice system might be any number of things simultaneously—academics, convenience, demographics—but quantifying the relative strength of parental preferences for these attributes provides a basis for making important and useful predictions about behavioral responses to policy changes.

Authors' Note

The views and conclusions expressed in this report are those of the authors and do not necessarily reflect those of the Walton Family Foundation.

Acknowledgments

The authors thank Kai Filipczak for expert programming assistance and Eric Isenberg and Kata Mihaly for reading drafts and making helpful suggestions. We thank Sujata Bhat, Catherine Peretti, and Aaron Parrott for patient and helpful guidance with D.C. choice data.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: The research was supported by a grant from the Walton Family Foundation.

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Manuscript received July 1, 2016

First revision received December 27, 2016

Second revision received February 24, 2017

Accepted February 27, 2017